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```
#include <OneWire.h>
#include <DallasTemperature.h>

// Data wire is plugged into port 2 on the Arduino
#define ONE_WIRE_BUS 14

// Setup a oneWire instance to communicate with any OneWire devices
OneWire oneWire(ONE_WIRE_BUS);

// Pass our oneWire reference to Dallas Temperature.
DallasTemperature sensors(&oneWire);

void setup(void) {
  // Start serial communication for debugging
  Serial.begin(115200);
  Serial.println("DS18B20 Temperature Sensor Test");

  // Start up the library
  sensors.begin();
}

void loop(void) {
  // Call sensors.requestTemperatures() to issue a global temperature
  // request to all devices on the bus
  Serial.print("Requesting temperatures...");
  sensors.requestTemperatures();
  Serial.println("DONE");

  float tempC = sensors.getTempCByIndex(0);

  // Check if reading was successful
  if(tempC != DEVICE_DISCONNECTED_C) {
    Serial.print("Temperature for the device 1 (index 0) is: ");
    Serial.print(tempC);
    Serial.println("°C");
  } else {
    Serial.pri... Could not read temperature data");
```

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```
  }

  delay(2000);
}


// #include <OneWire.h>

// OneWire ds(2); // Sensor on digital pin 2

// void setup() {
//   Serial.begin(115200);
// }

// void loop() {
//   byte addr[8];

//   if (!ds.search(addr)) {
//     Serial.println("No more addresses. Searching
again...");
//     ds.reset_search();
//     delay(2000);
//     return;
//   }

//   Serial.print("ROM =");
//   for(int i = 0; i < 8; i++) {
//     Serial.write(' ');
//     Serial.print(addr[i], HEX);
//   }

//   if (OneWire::crc8(addr, 7) != addr[7]) {
//       Serial.println(" - CRC is not valid!");
//       return;
//   }
//   Serial.println(" - SUCCESS: Sensor found!");
//   delay(2000);
// }  Error: Could not read temperature data
Requesting temperatures...DONE   whatb is thereasonn
for this error
```

That error message:

```
Error: Could not read temperature data
```

means your sensor is returning `DEVICE_DISCONNECTED_C`, which usually points to a **hardware or communication issue**, not a bug in your code.

Let's break down the most common reasons 👇

🔴 1. Wrong wiring (MOST common)

The DS18B20 has **3 pins**, and incorrect wiring will cause exactly this error.

Correct wiring:

- **VCC → 3.3V or 5V**
- **GND → GND**
- **DATA → GPIO 14** (since you used `#define ONE_WIRE_BUS 14`)
- ⚠️ **Important:** Add a **4.7kΩ pull-up resistor** between **DATA and VCC**

👉 Without the resistor, the sensor often shows *"Could not read temperature data"*

🔴 2. Wrong GPIO pin

You defined:

```cpp
#define ONE_WIRE_BUS 14
```

But your commented code uses:

```cpp
OneWire ds(2);
```

👉 Make sure:

- Your sensor DATA wire is actually connected to **GPIO 14**
- Or change code to match your wiring:

```cpp
#define ONE_WIRE_BUS 2
```

🔴 3. Sensor not detected at all

Your second (commented) code is a **scanner**. That's good 👍

👉 Run that code:

- If it prints **"No more addresses" repeatedly**, your sensor is **not detected**
- If it finds an address → wiring is OK

🔴 4. Power issue (especially with ESP8266/ESP32)

If you're using:

- ESP8266 or ESP32

Then:

- Try **3.3V instead of 5V**
- Ensure stable power (USB ports can be weak sometimes)

🔴 5. Faulty sensor

If everything is correct:

- Wiring ✅
- Resistor ✅
- Correct pin ✅

👉 Then the sensor itself might be damaged.

🔴 6. Loose connections / breadboard issues

Very common in practice:

- Loose jumper wires
- Bad breadboard rows

✅ Quick checklist

✔ DATA → correct GPIO (14?)
✔ 4.7k resistor between DATA & VCC
✔ VCC & GND correct
✔ Try address scanner code
✔ Try another GPIO pin
✔ Try another sensor

💡 Debug tip

Add this to confirm detection:

```cpp
Serial.println(sensors.getDeviceCount());
```

👉 If it prints `0`, your sensor is not being detected at all.

If you want, send me:

- Your board type (Arduino Uno / ESP8266 / ESP32)
- A photo or wiring description

I can pinpoint the exact issue for you 👍

esp32

Got it—you're using an ESP32. That narrows things down a lot 👍

Your error is **very common on ESP32 + DS18B20**, and it's almost always one of these:

🔴 Most Likely Issues (ESP32-specific)

✅ 1. Missing pull-up resistor (VERY IMPORTANT)

ESP32 **will NOT work reliably without it**

👉 You must connect:

- **4.7kΩ resistor between DATA and 3.3V**

Without this:

- Sensor may be detected sometimes
- But temperature read fails → exactly your error

✅ 2. Wrong voltage (don't use 5V blindly)

ESP32 is **3.3V logic**

👉 Best setup:

- VCC → **3.3V**
- DATA → GPIO 14
- GND → GND

⚠️ Using 5V can sometimes cause unstable readings or even damage over time.

✅ 3. GPIO 14 is OK, but check wiring carefully

GPIO 14 works fine 👍
But ESP32 pins can be tricky.

👉 Try switching to a safer pin:

```cpp
#define ONE_WIRE_BUS 4
```

(GPIO 4 is very commonly used and stable)

✅ 4. Sensor not detected (test this)

Add this in `setup()` :

```cpp
Serial.print("Device count: ");
Serial.println(sensors.getDeviceCount());
```

👉 Result:

- `0` → wiring problem ❌
- `1` → sensor detected ✅ (then issue is communication)

✅ 5. Timing issue (ESP32 sometimes needs delay)

Try adding a small delay after requesting:

```cpp
sensors.requestTemperatures();
delay(750);  // important for ESP32 sometimes
float tempC = sensors.getTempCByIndex(0);
```

✅ 6. Check if you're getting -127°C

If you print:

```cpp
Serial.println(tempC);
```

👉 If you see:

```
-127.00
```

That means:
👉 **Sensor not connected / not responding**

🔧 Recommended Working Setup (ESP32)

Wiring:

- VCC → 3.3V
- GND → GND
- DATA → GPIO 4 (recommended)
- 4.7k resistor → DATA ↔ 3.3V

Code fix:

```cpp
```

```
#define ONE_WIRE_BUS 4

void loop(void) {
  sensors.requestTemperatures();
  delay(750);  // important

  float tempC = sensors.getTempCByIndex(0);

  if(tempC != DEVICE_DISCONNECTED_C) {
    Serial.print("Temp: ");
    Serial.println(tempC);
  } else {
    Serial.println("Error: Could not read temperature data");
  }

  delay(2000);
}
```

🚀 Quick diagnosis shortcut

Answer these and I'll pinpoint it instantly:

1. Did you use a **4.7k resistor**?
2. What pin is DATA connected to?
3. What does `getDeviceCount()` print?

If I had to guess based on your error 👉
💥 **95% chance: missing pull-up resistor**